



BUSINESS INCOME UNDERREPORTING AND PUBLIC FINANCE

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The Macroeconomics of US Public Finance

- Net government saving ≈ -1 \$Trillion (in 2018)
 - Current receipts: 5.6T
 - Current expenditures: 6.7T



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 - Current receipts: 5.6T
 - Current expenditures: 6.7T
- Untaxed business income ≈ 1 \$Trillion
 - Income reported to IRS: 3.2T
 - Estimate of true: 4.2T



The Macroeconomics of US Public Finance

- Net government saving ≈ -1 \$Trillion
 - Current receipts: 5.6T
 - Current expenditures: 6.7T
- Untaxed pass-through income ≈ 700 \$Billion
 - Income reported to IRS: 1.3T
 - Estimate of true: 2T



The Macroeconomics of US Public Finance

- Net government saving $\approx -5.4\%$ GDP
 - Current receipts: 27%
 - Current expenditures: 33%
- Untaxed pass-through income $\approx 3.4\%$ GDP
 - Income reported to IRS: 6.4%
 - Estimate of true: 9.8%



The Macroeconomics of US Public Finance

- Net government saving $\approx -5.4\%$ GDP
 - Current receipts: 27%
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- Untaxed pass-through income $\approx 3.4\%$ GDP
 - Income reported to IRS: 6.4%
 - Estimate of true: 9.8%

$\Rightarrow$ Prompting more funding for IRS enforcement



Greater IRS Enforcement

- Inflation Reduction Act:
 - 80 billion over 10 years
 - Enforcement budget roughly doubled
- Predicted returns on investment (ROI):
 - CBO/JCT (2021): 5–9%
 - Boning et al (2023): 5–12%



This Paper

- First step before using IRS micro data
 - Use public IRS compliance data (TCMP/NRP)
 - Develop dynamic GE model of tax evasion
 - Compare higher tax vs enforcement counterfactuals
- Useful for next steps
 - Data: expand collection to business filings
 - Theory: add transition dynamics and welfare analysis



What's New?

- Factors relevant for *dynamics of tax evasion*
 - Loss of *sweat capital* (eg, reputation, brands, etc)
 - Recovery of *back taxes*
 - Why relevant?
 - Impacts business dynamics and productivity
 - Amplifies precautionary motives
- ⇒ Economies with higher tax vs enforcement different



Summary of IRS Compliance Data



IRS Compliance Data

- Tax gap = random audits + DCE adjustments
- Random audits:
 - Taxpayer compliance measurement program, 1962–88
 - National research program, 2000–present
- *Detection controlled estimation* (DCE) adjustments:
 - Scale up recommendations of all examiners
 - Use data from examiners with largest adjustments



How Big is the Tax Gap?

Gross tax gap	2001	2011	2021
Amount:			
billions of 2023\$	567	575	763
% of GDP	3.3	2.7	2.9
Source share (%):			
Nonfiling	7	8	11
Underpayment	10	12	10
Underreporting	83	80	80
Business income	52	44	44
Wages & salaries	3	3	2
Other	28	34	35



How Widespread is Cheating?

- Evidence from 2 NRP random-audit studies (no DCE)
 - All owners—ranked by reported incomes
 - Sole proprietors—ranked by understated tax
- Reveal same patterns
 - Cheating is widespread
 - But few owners account for most cheating



Does Evasion Occur Across Income Distribution?

- Evidence from NRP random-audit studies:
 - Underreporting occurs across the distribution
 - Top-income estimates depend on DCE adjustments
- ⇒ Wide range of opinions on enforcement targeting



Is Economic Deterrence Driving Compliance?

- TAS surveys intended to elicit nonpecuniary motives
- How?
 - Construct samples of sole proprietors
 - Use DIF scores indicating likelihood of audit
 - Group proprietors by DIF score
 - Compare responses of low- and high-DIF groups
- What do they find?
 - Trust in IRS/govt is main predictor of compliance



Recap: Lessons from IRS Data

- Gross tax gap large and $\approx 3\%$ of GDP over time
- Underreporting is main source of tax gap
- Underreporting by business owners is most of that
- Underreporting is widespread but concentrated
- Underreporting occurs across the income distribution
- Economic deterrence is only one factor driving compliance



Theory



Key Factors

- Occupational choice: paid- or self-employment
- Taxpayer types: always compliant or not
- Noncompliance source: business income underreporting
- Dynamics of tax evasion:
 - Loss of reputation, business brands, customers
 - Recovery of back taxes

⇒ Extends standard model of economic deterrence



Occupational Choice

- Choose business b or work w

$$V(s) = \max \{V^b(s), V^w(s)\}$$

$$V^i(s) = \max_x \{U(c, \ell) + \beta \sum_{z', \epsilon'} \pi(z', \epsilon' | z, \epsilon) \mathcal{V}(s')\}$$

where $s = (a, \kappa, d, z, \epsilon)$ and

- a : financial assets
- κ : *sweat capital*, eg, reputation, brands, etc
- d : *back taxes*, eg, accumulated unpaid taxes
- z : productivity in self-employment
- ϵ : productivity in paid-employment
- $x = [a', \kappa', d', \underbrace{c_p, c_c, \ell, k_p, h_p, n_p, h_\kappa, e, c^r, y_b^r}_{\text{decisions defined below}}]$



Continuation Value

$$\begin{aligned}\mathcal{V}(a', \kappa', d', z', \epsilon') \\&= \underbrace{(1 - \Pi(d'))V(a', \kappa', d', z', \epsilon')}_{\text{no audit}} \\&\quad + \underbrace{\Pi(d')V(a' - f_a(d'), f_r(\kappa'), 0, z', \epsilon')}_{\text{audit}}\end{aligned}$$



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↑ ↗

Probability of audit and fines depend on d'



Continuation Value

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$$+ \underbrace{\Pi(d')V(a' - f_a(d'), f_r(\kappa'), 0, z', \epsilon')}_{\text{audit}} \uparrow$$

Audit results in reputational losses



Business Owner's Technologies

- Goods and services: $y_p = z f_p(\kappa, h_p, k_p, n_p)$
 - z = productivity in self-employment
 - κ = sweat capital
 - h_p = owner time in production
 - k_p = rented physical capital
 - n_p = employee time
- Sweat investment: $x_\kappa = f_\kappa(h_\kappa, e)$
 - h_κ = owner time in brand building
 - e = owner expenses



Business Owner's Constraints

- Budget

$$a' = [(1+r)a + y_b - T^b(y_b^r) - (1+\tau_c)(c_c + pc_p) + \chi] / (1+\gamma)$$

- Sweat capital

$$\kappa' = [(1 - \delta_\kappa)\kappa + f_\kappa(h_\kappa, e)] / (1 + \gamma)$$

- Back taxes

$$d' = [(1 - \delta_d)d + f_d(c^r)] / (1 + \gamma)$$

- Borrowing

$$a' \geq f_a(d')$$



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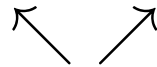
next period and current assets



Business Owner's Constraints

- Budget

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true and reported income

$$y_b = py_p - (r + \delta)k_p - wn_p - e$$

$$y_b^r = y_b - (1 + \tau_c)c^r$$



Business Owner's Constraints

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taxes on business and consumption



Business Owner's Constraints

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↑ ↗

goods produced by C-corps
and pass-thrus, $c = \text{ces}(c_c, c_p)$



Business Owner's Constraints

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↑

transfers



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↑

sweat investment (shown earlier)



Business Owner's Constraints

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↑

current misreporting: $y_b - y_b^r = (1 + \tau_c)c^r$



Business Owner's Constraints

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$$a' = [(1+r)a + y_b - T^b(y_b^r) - (1+\tau_c)(c_c + pc_p) + \chi] / (1+\gamma)$$

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- Borrowing

$$a' \geq f_a(d')$$



Close the Model

- Standard dynamic program for workers, except
 - Sweat capital decays without use
 - Back taxes not forgiven
- Standard dynamic program for C corporations
- Public financing (G&S plus transfers) with
 - Taxes on consumption and all forms of income
 - Fines if caught evading taxes

(Details in slide deck appendix)



Qualitative Predictions

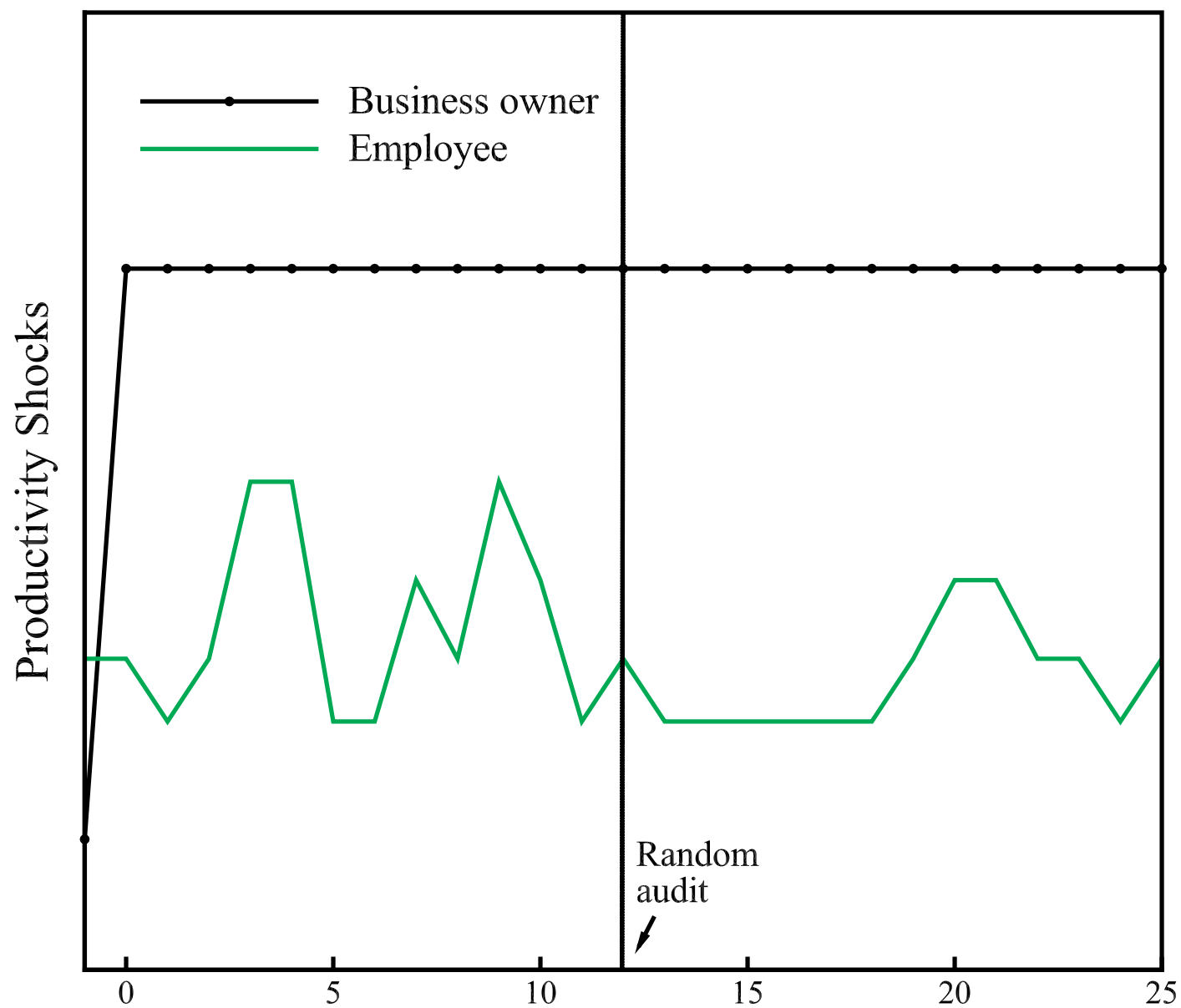


To Highlight Key Mechanisms...

- Consider four alternative economies with
 1. Low audit rate & no loss of reputation with audit
 2. Low audit rate & loss of reputation with audit
 3. High audit rate & no loss of reputation with audit
 4. High audit rate & loss of reputation with audit
- In each, track *the same* business owner over time

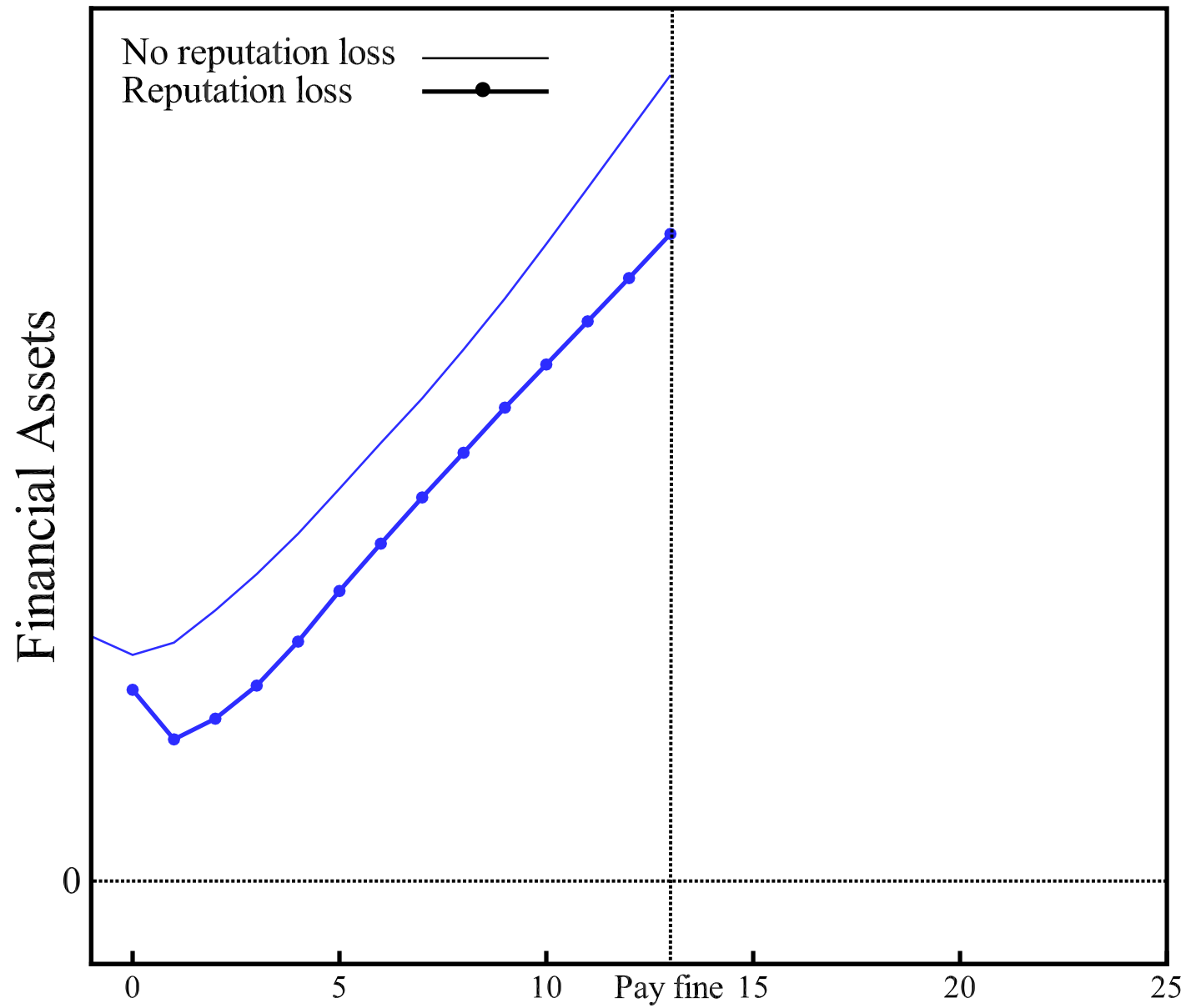


The Shocks



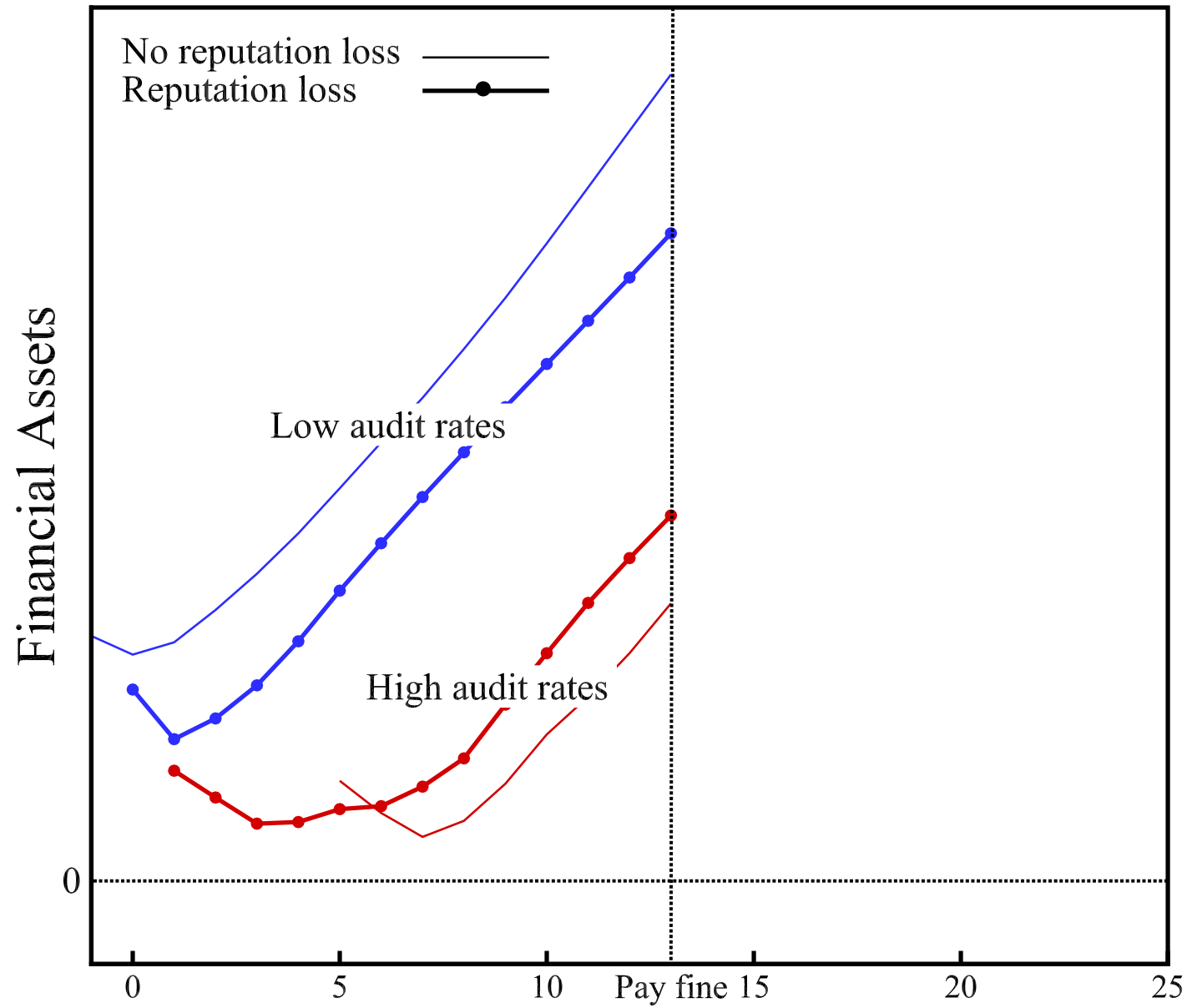


Financial Assets Before Fines Due



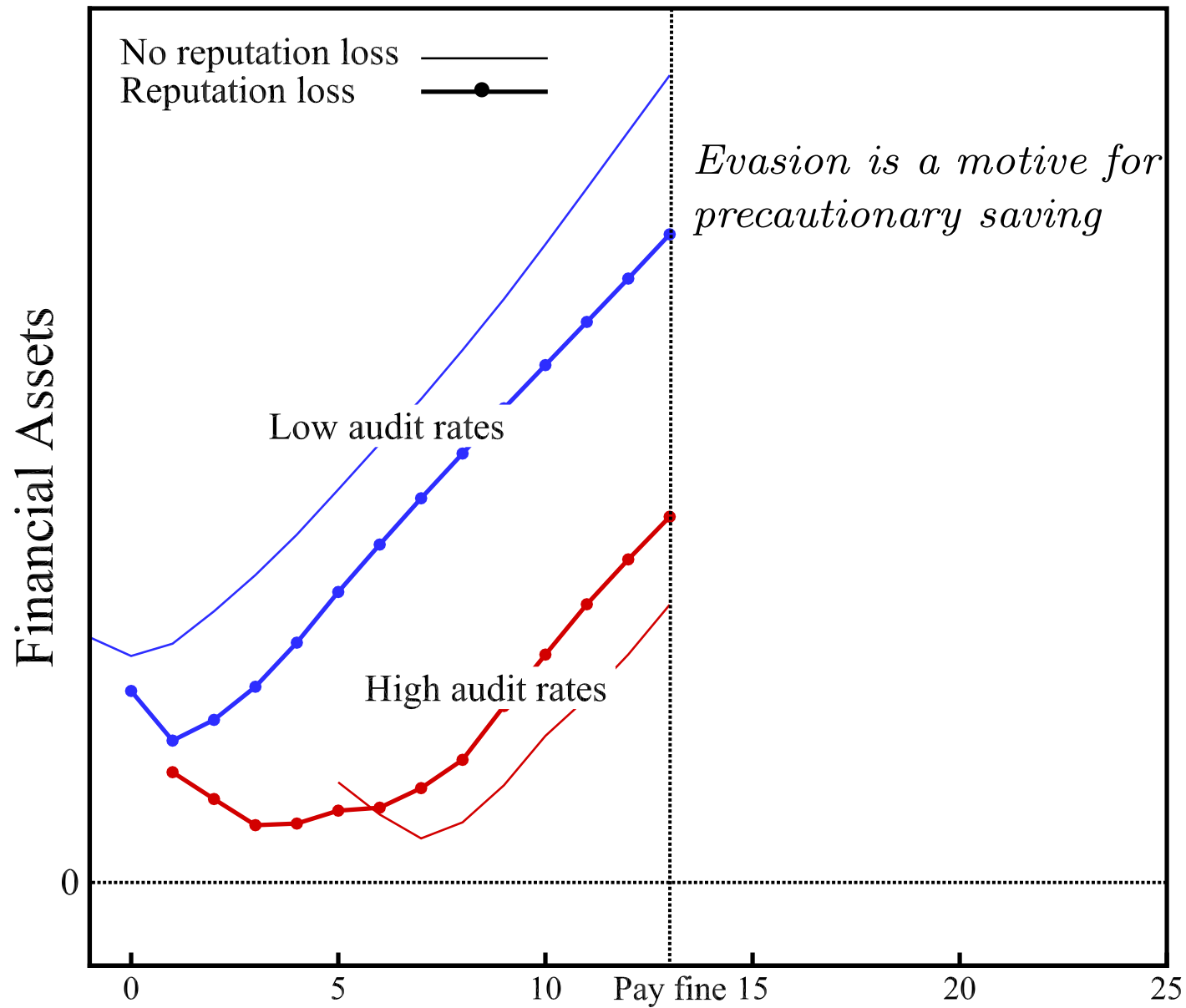


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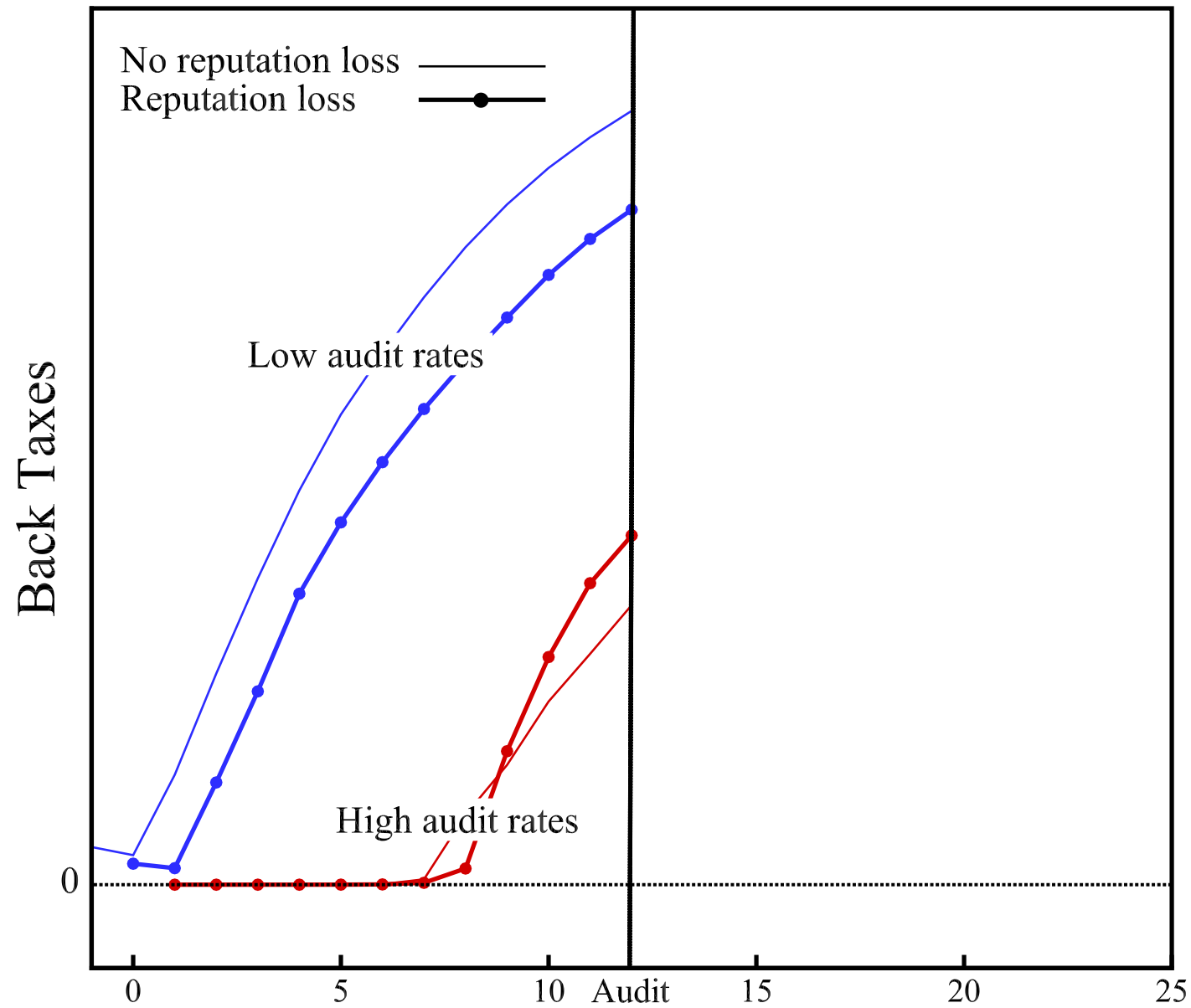


Financial Assets Before Fines Due



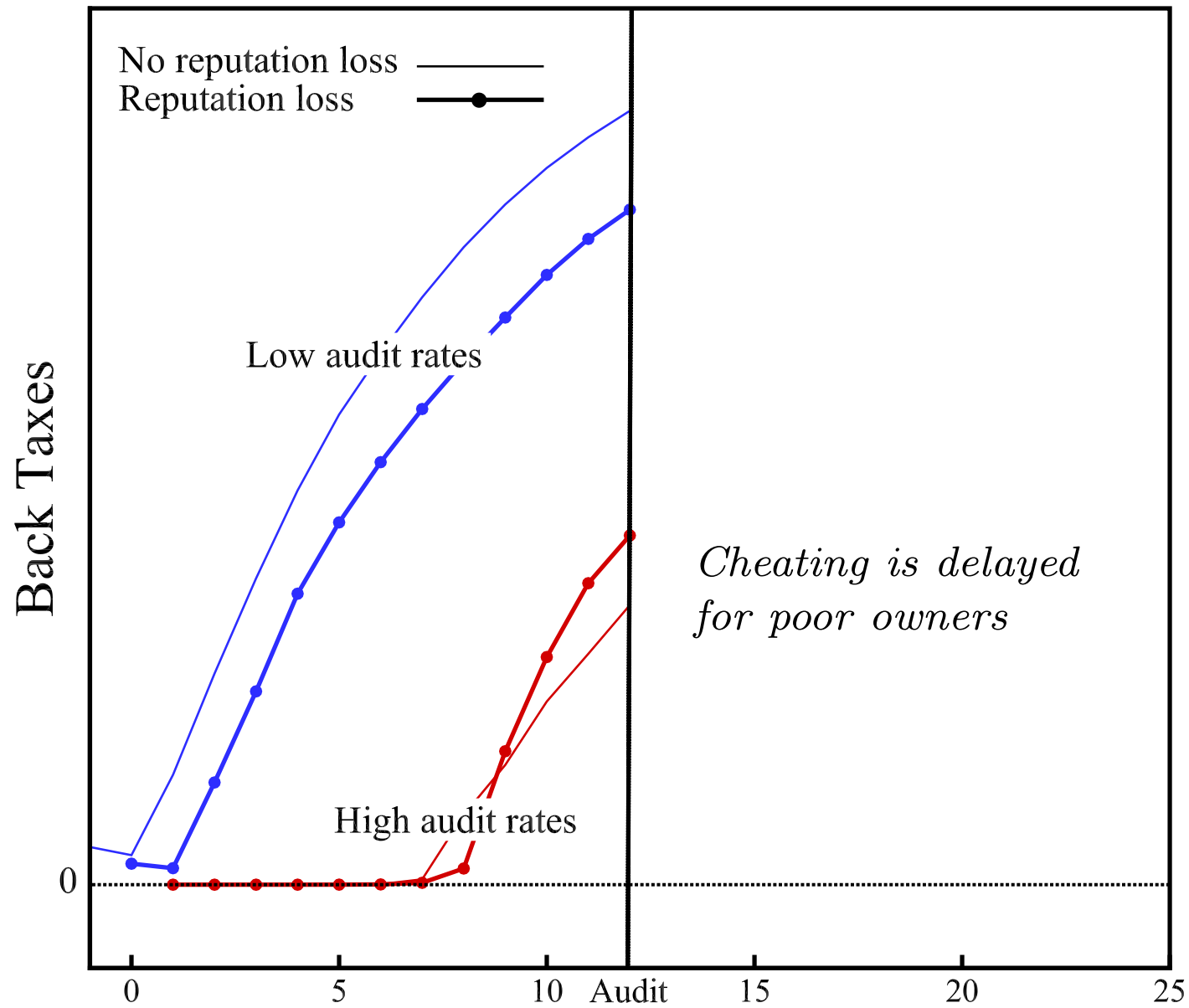


Back Taxes Before Audit



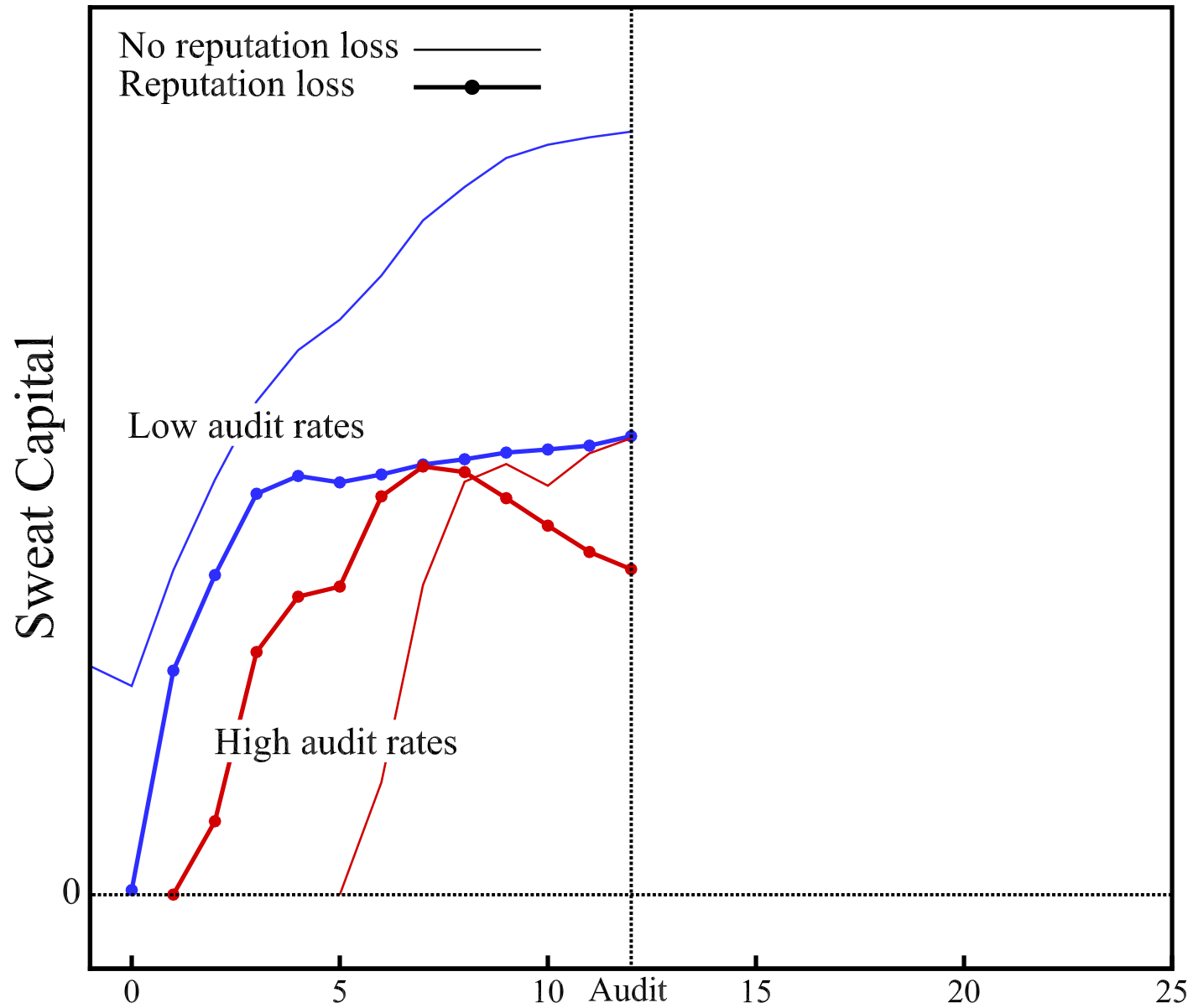


Back Taxes Before Audit



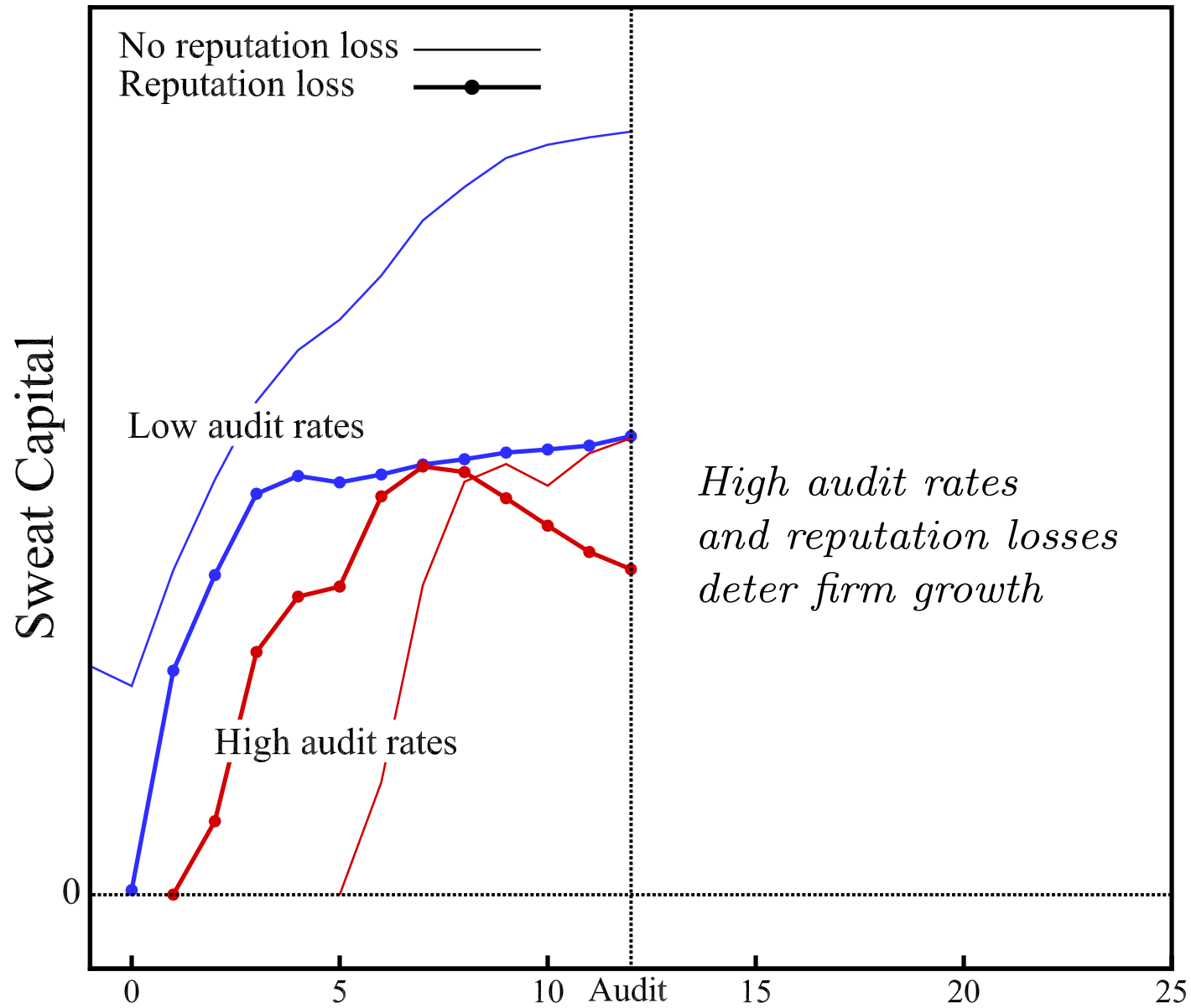


Sweat Capital Before Audit



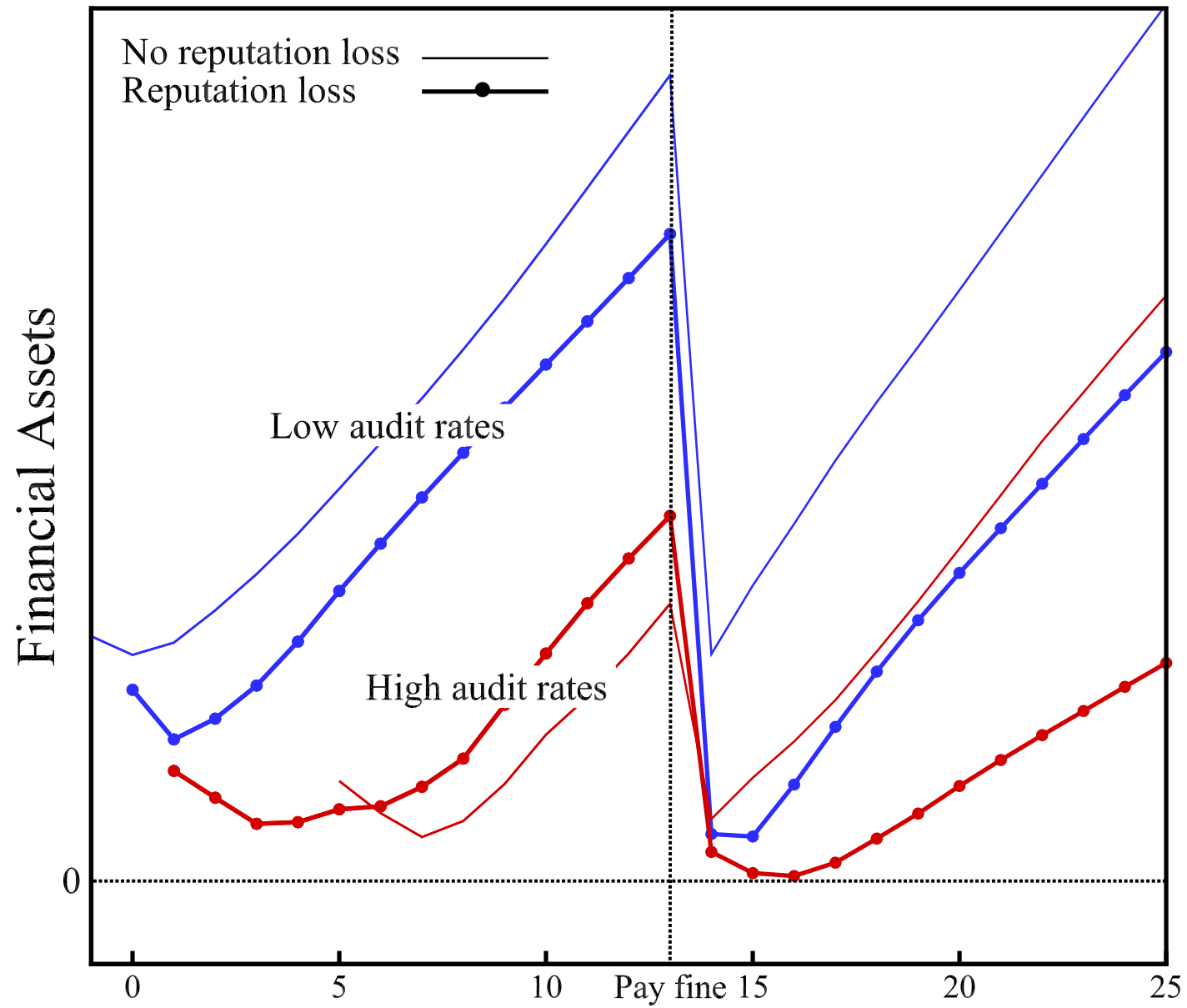


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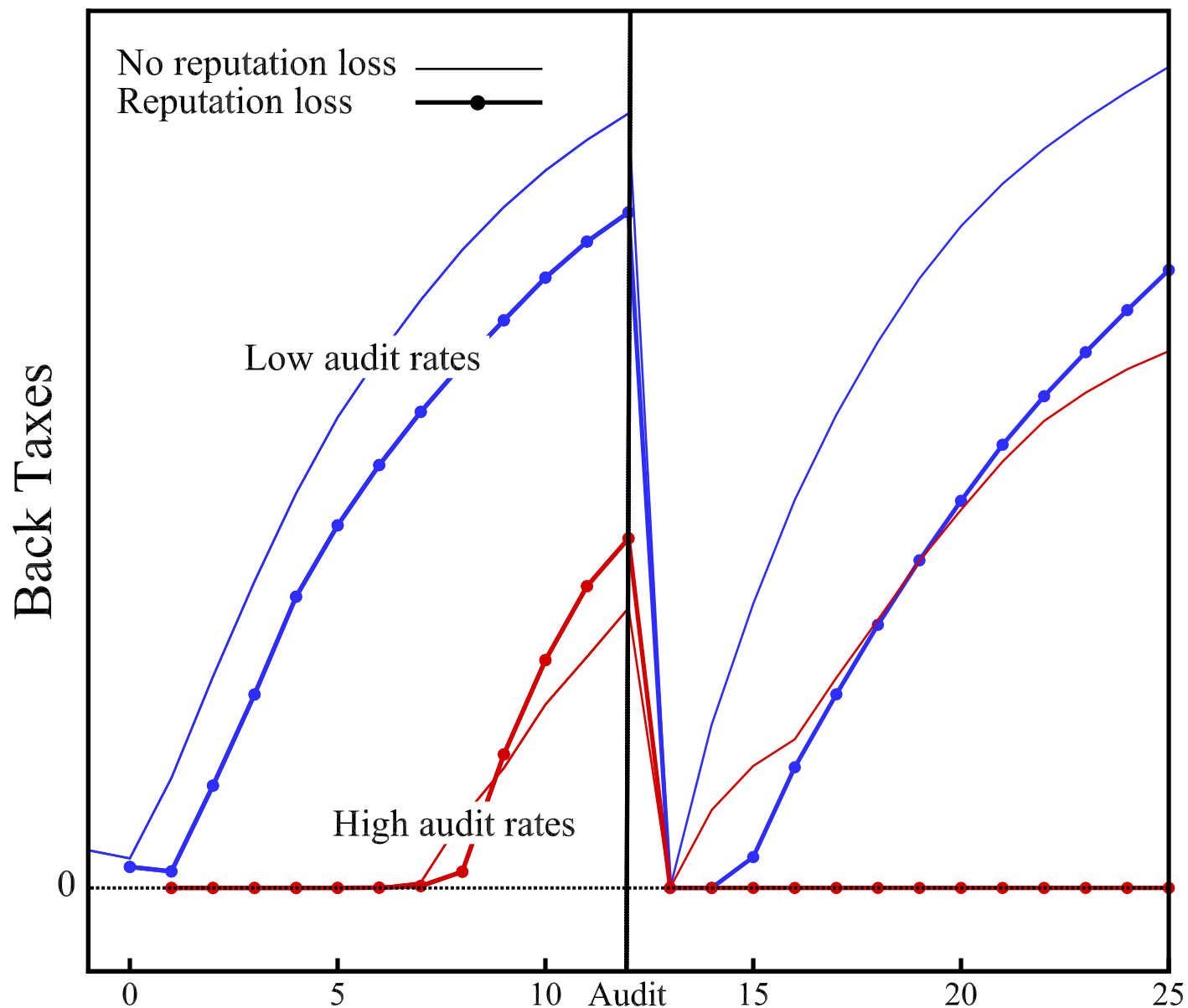


Financial Assets After Audit



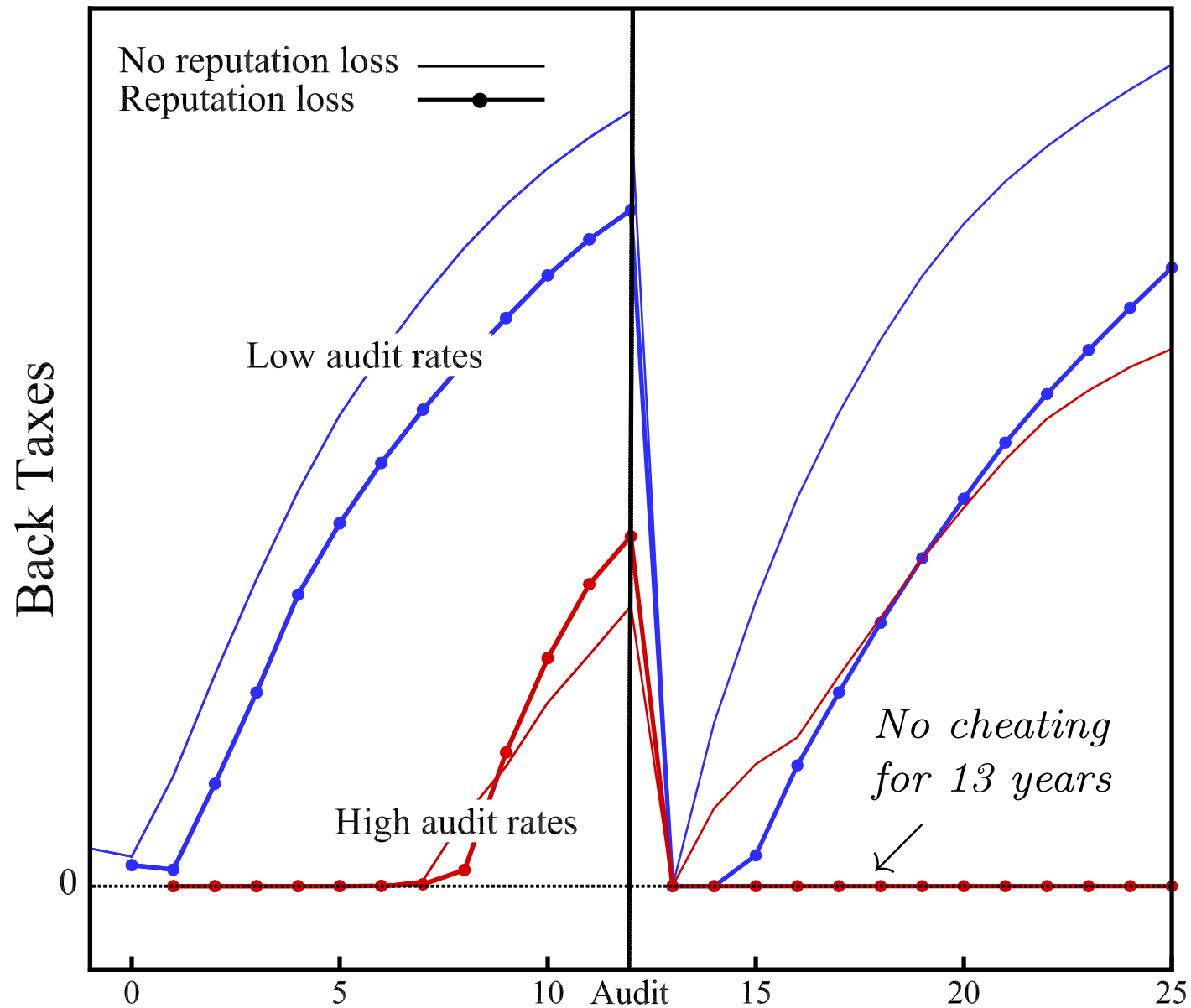


Back Taxes After Audit



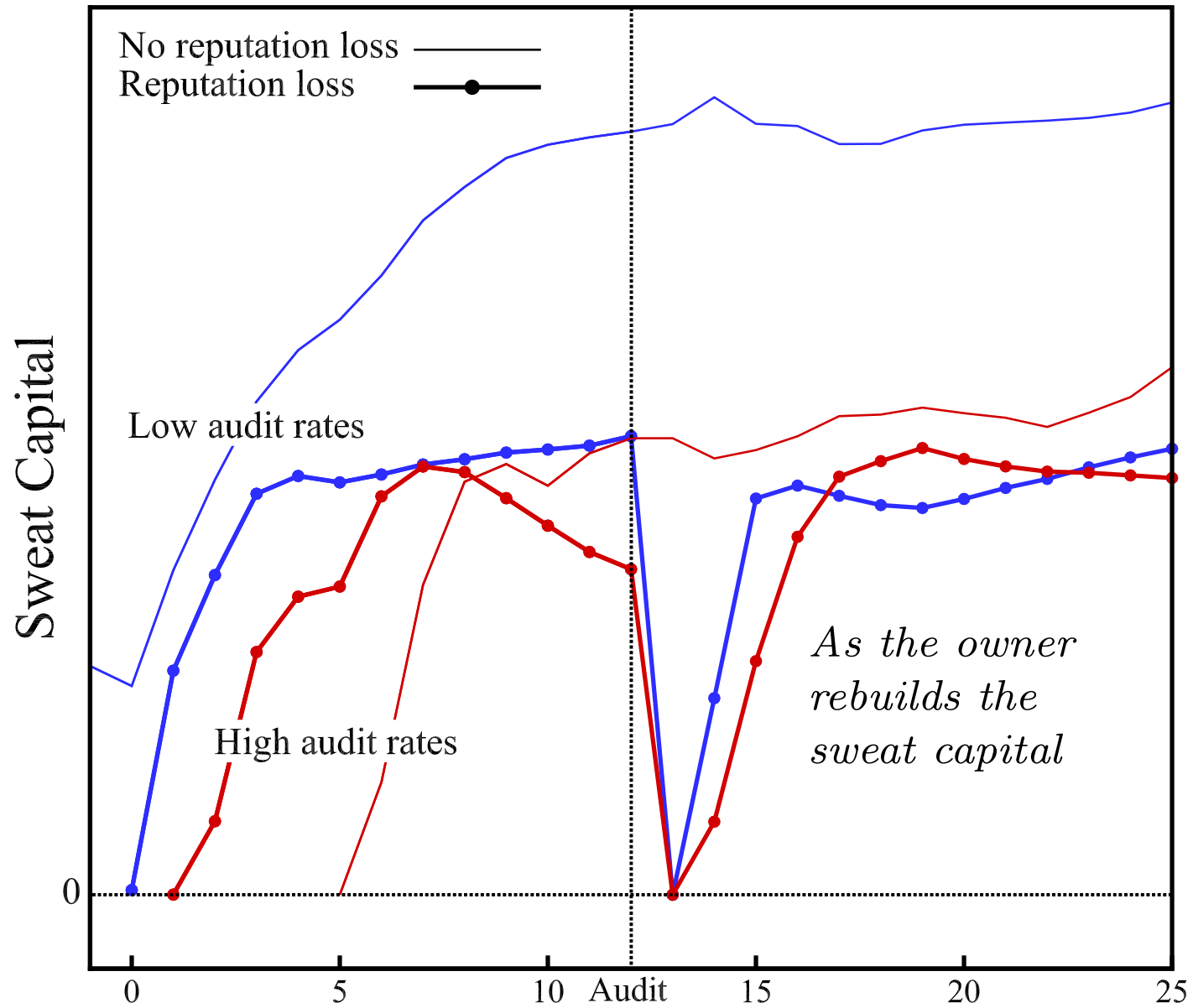


Back Taxes After Audit





Sweat Capital After Audit





Predictions of Increased Enforcement

- Lower precautionary motives
 - Financial assets used to pay future fines
 - Borrowing constraints less binding
- Lower sweat capital stocks
 - Brand assets lost with exposed tax evasion
 - Business ages lower with more exit/entry
 - Business productivity higher due to selection



Quantitative Results



Key Compliance Parameters

- Audit probability, $\Pi(d') = \pi$, π varied
- Fines, $f_a(d) = \bar{p}d$, $\bar{p} = 4$
- Reputational cost, $f_r(\kappa) = 0$ if non-compliant
- Underreporting, $f_d(c^r) = \tau_b(1 + \tau_c)c^r$, $\tau_c = .065$, $\tau_b = .4$
- Back taxes depreciation, $\delta_d = 20\%$



Comparative Statics

- Vary audit probability π
- Record impacts for owners by type



Fraction of Population

Owner type:	% Change from $\pi = 2\%$ to			
	3%	4%	5%	6%
Non-compliant	−23	−40	−63	−61
Compliant	11	20	30	26
All owners	−3	−6	−10	−11

⇒ Large compositional shift



Financial Assets (*a*)

Owner type:	% Change from $\pi = 2\%$ to			
	3%	4%	5%	6%
Non-compliant	−20	−36	−55	−65
Compliant	8	21	36	36
All owners	−18	−26	−30	−33

⇒ Large drop in precautionary saving



Productivity (z)

Owner type:	% Change from $\pi = 2\%$ to			
	3%	4%	5%	6%
Non-compliant	8	16	19	18
Compliant	-1	-2	-3	-3
All owners	3	5	4	4

$\Rightarrow$ Increase in productivity due to selection



Sweat Capital (κ)

Owner type:	% Change from $\pi = 2\%$ to			
	3%	4%	5%	6%
Non-compliant	-12	-25	-25	-4
Compliant	3	7	13	13
All owners	-1	0	8	11

$\Rightarrow$ Increase in capital with more compliance



Business Age

Owner type:	% Change from $\pi = 2\%$ to			
	3%	4%	5%	6%
Non-compliant	-46	-63	-71	-66
Compliant	5	12	22	20
All owners	-27	-32	-27	-27

⇒ Large drop in age with increased auditing



Summary: All Owners

Owner type:	% Change from $\pi = 2\%$ to			
	3%	4%	5%	6%
% of population	−3	−6	−10	−11
Business age	−27	−32	−27	−27
Financial assets	−18	−26	−30	−33
Productivity	3	5	4	4
Sweat capital	−1	0	8	11

⇒ Large drops in counts, age, financial assets

But increased TFP due to selection



Policy Counterfactuals

- Two ways to raise same revenues
 - Higher enforcement: $\pi = 2\% \rightarrow \pi = 5\%$
 - Higher tax rate on business: $\tau_b = 40\% \rightarrow \tau_b = 43\%$
- Raise revenues by 6% relative to $\pi = 2, \tau_b = 40$ baseline



Enforcement vs Taxation

% Change in:	More Audits $\pi=5$ vs 2%	Higher Tax $\tau_b=43$ vs 40%
# of Owners	-10	-5
Non-compliant	-63	6
Compliant	30	-13
Back taxes, d	-76	20
Business age	-27	22
Financial assets, a	-30	14
Sweat capital, κ	8	9
Productivity, z	4	-2
Business income, y_b	6	2



Bottom Line

- Higher enforcement vs taxation
 - Most evident in composition of businesses/owners
 - Less evident in aggregate business activity
- Need transitional dynamics to do proper welfare analysis



Next Steps

- Data: gather relevant IRS micro data
 - Current NRP studies only work with 1040
 - Want to expand analysis to business filings
- Theory: add transitional dynamics
 - Current analysis is steady state
 - Want to analyze Inflation Reduction provisions
 - Want to do full welfare analysis with transition



Appendix



Dynamic Program for Workers

- Workers choose $x = [a', c_p, c_c, \ell]$ to solve

$$V^w(s) = \max_x \{U(c, \ell) + \beta \sum_{z', \epsilon'} \pi(z', \epsilon' | z, \epsilon) \mathcal{V}(s')\}$$

subject to

$$a' = [(1 + r)a + w\epsilon h_w - T^w(w\epsilon h_w) - (1 + \tau_c)(c_c + pc_p) + \chi]/(1 + \gamma)$$

$$\kappa' = (1 - \lambda_\kappa)\kappa/(1 + \gamma)$$

$$d' = (1 - \lambda_d)d/(1 + \gamma)$$

$$1 = \ell + h_w$$



Dynamic Program for Corporations

- Corporations choose x_c, n_c to solve

$$V^c(k_c) = \max \left\{ (1 - \tau_d)d_c + \frac{1 + \gamma}{1 + r} V^c(k'_c) \right\}$$

subject to

$$d_c = AF(k_c, n_c) - wn_c - x_c - \tau_p(y_c - wn_c - \delta_k k_c)$$

$$x_c = (1 + \gamma)k'_c - (1 - \delta_k)k_c$$



Government Budget Constraint

$$\begin{aligned} g + \chi + (r - \gamma)b = & \tau_c \int (c_{ci} + pc_{pi}) di + \\ & + \tau_d(y_c - wn_c - (\gamma + \delta_k)k_c - \tau_p(y_c - wn_c - \delta_k k_c)) \\ & + \tau_p(y_c - wn_c - \delta_k k_c) + \int T^n(w\epsilon_i n_i) di \\ & + \int T^b(y_{ri}^b) di + \int \mathbf{1}_i f_a(d_i) di \end{aligned}$$



IRS Compliance Data



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- Tax gap = random audits + DCE adjustments
- Random audits:
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How Big is the Tax Gap?

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What is the Main Source of the Gap?

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Source share:			
Underreporting	83	80	80
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What is the Main Source of Underreporting?

Source share	2001	2011	2021
Business	62	55	55
Wages & salaries	4	3	2
Other	34	42	43

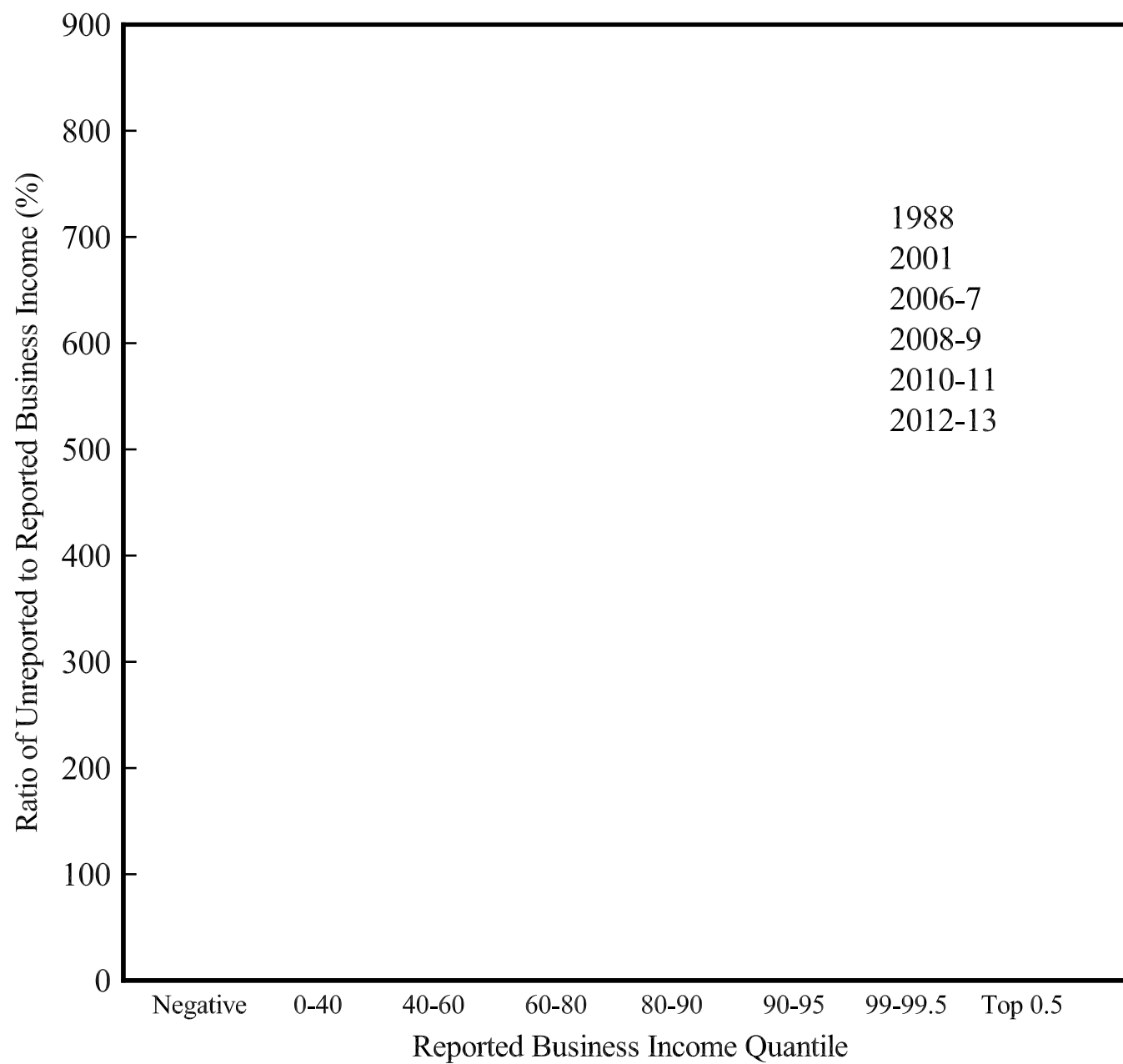


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- Evidence from 2 NRP random-audit studies (no DCE)
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- Reveal same patterns
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 - Few owners account for most cheating

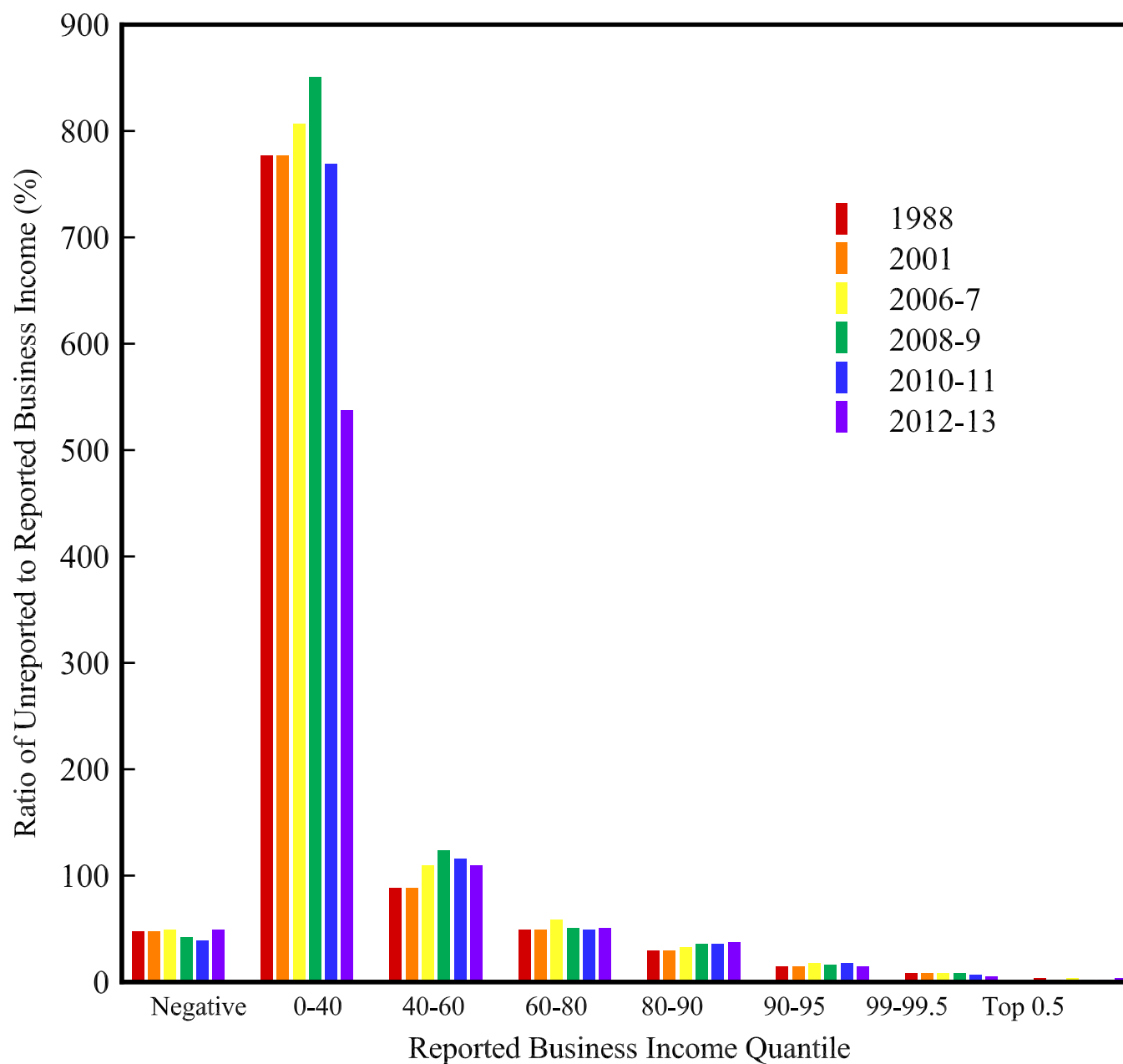


Owners Ranked by Reported Incomes



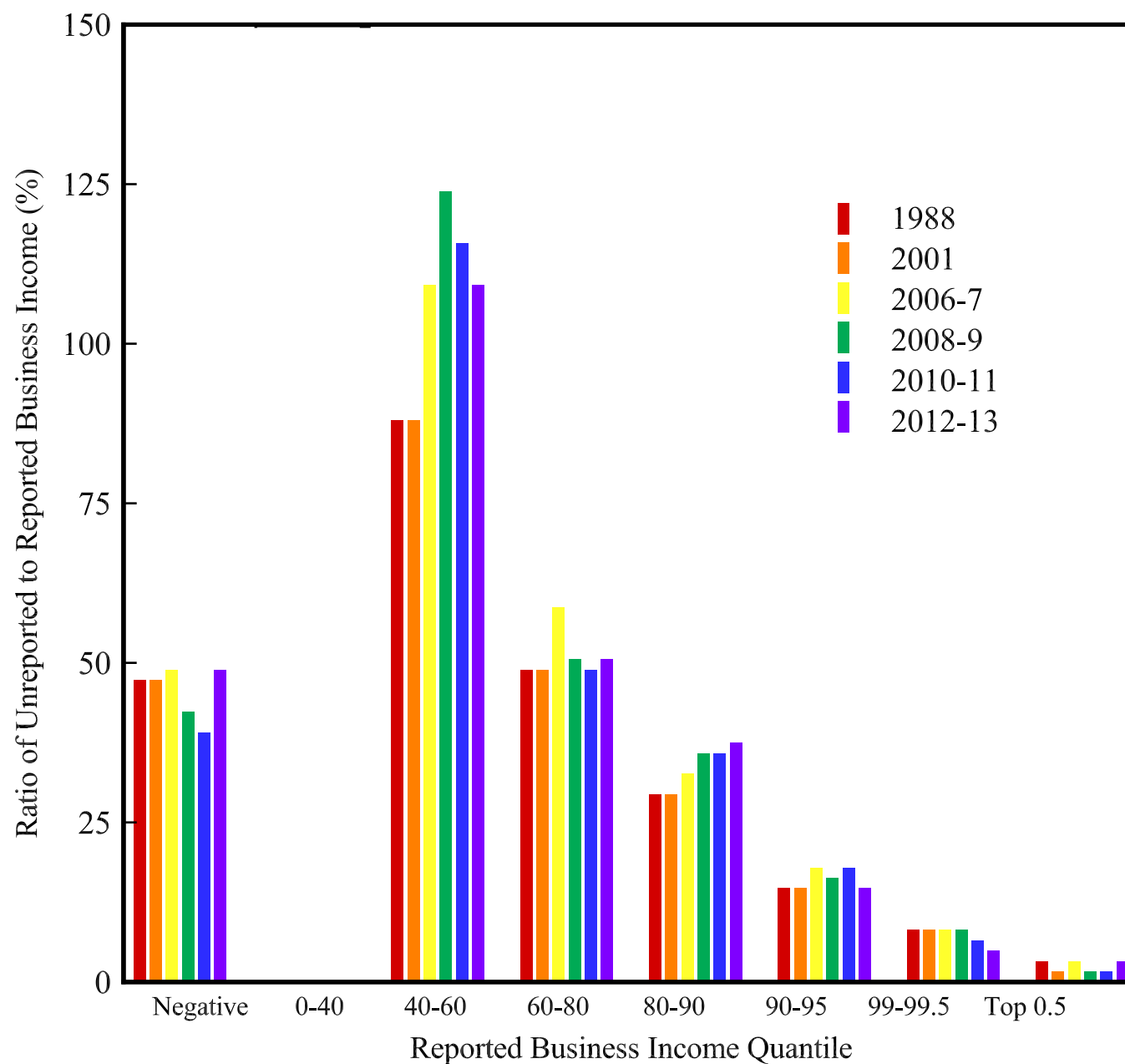


Owners Ranked by Reported Incomes



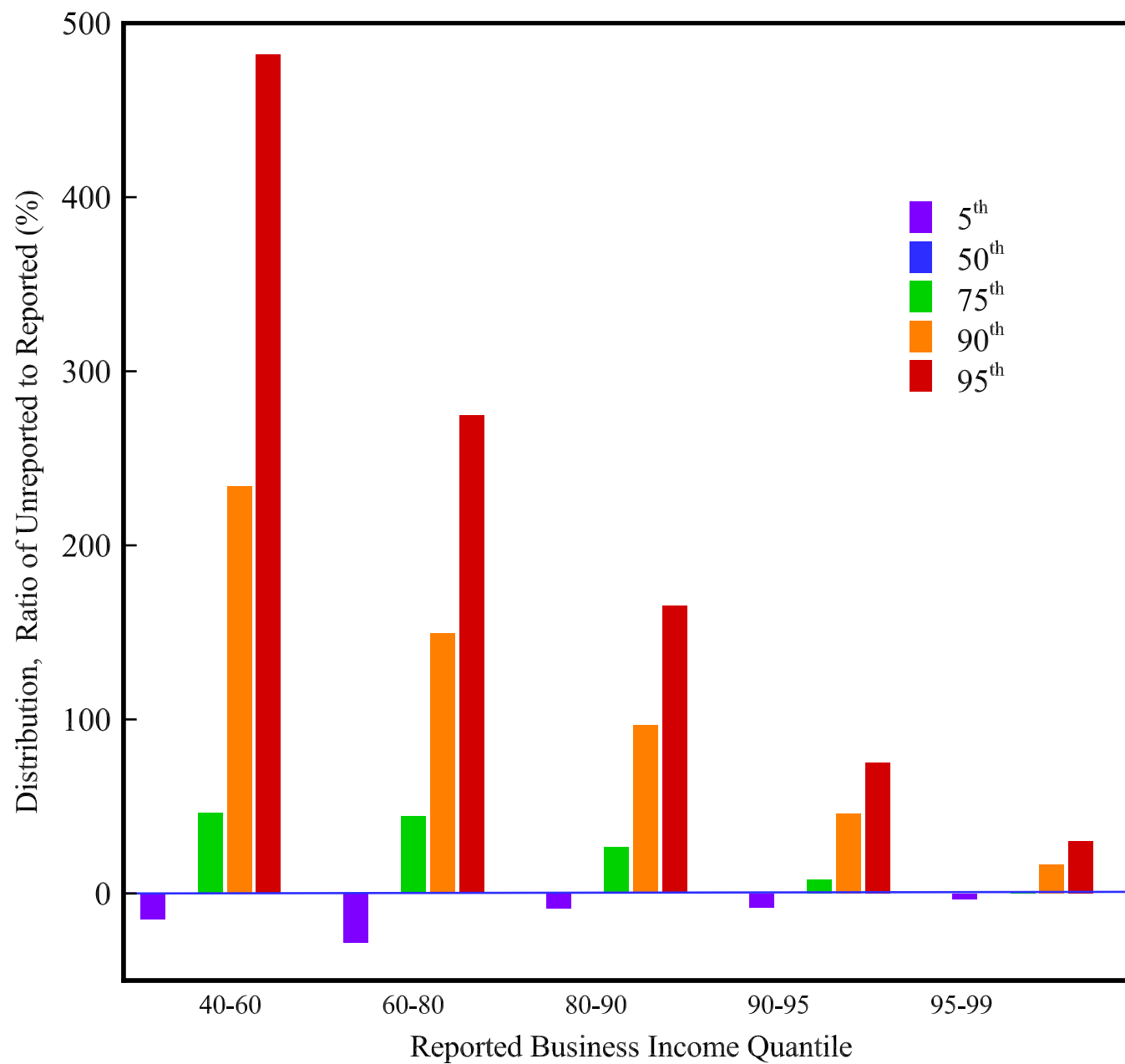


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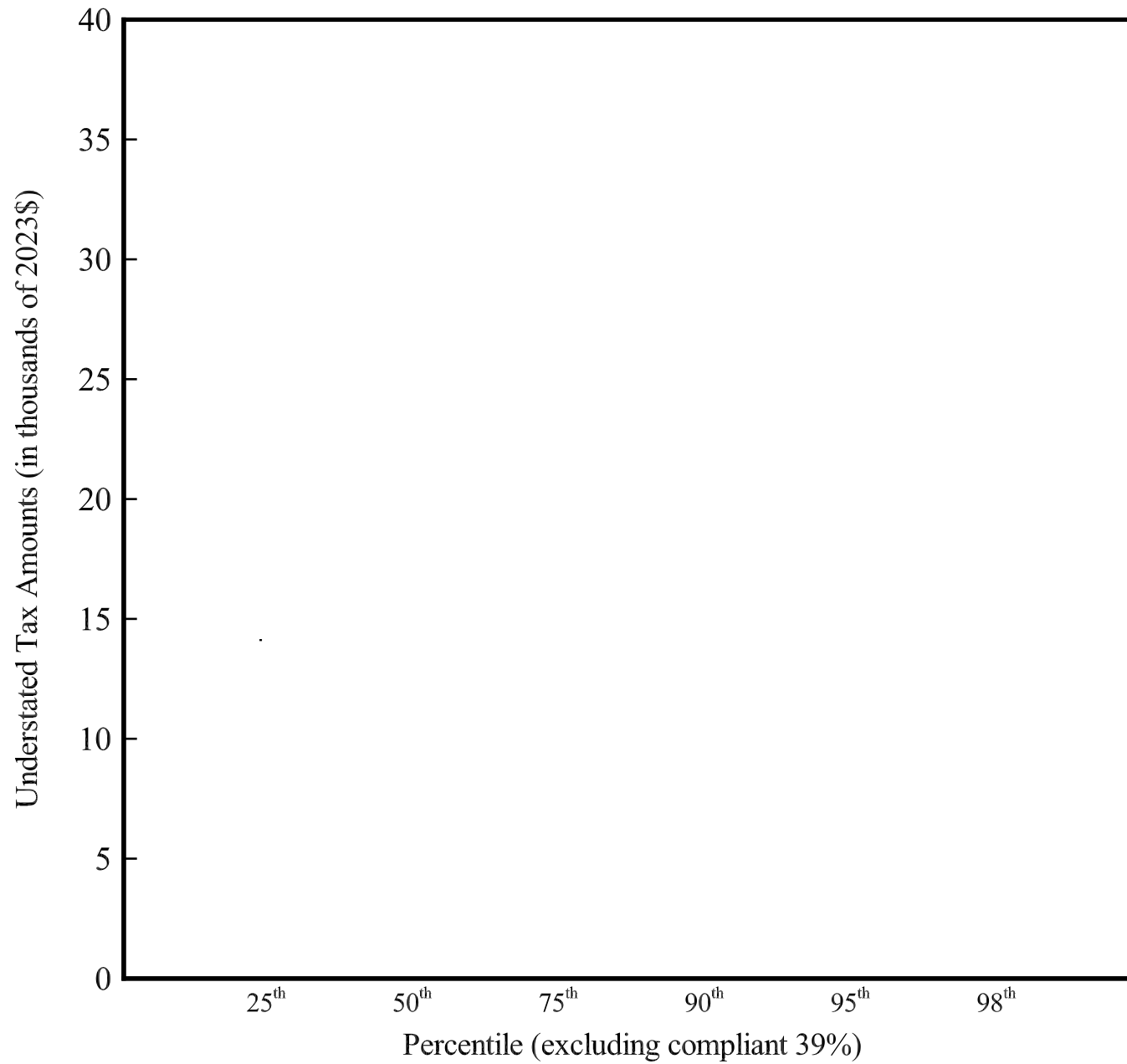


Distribution of U-to-R Ratios



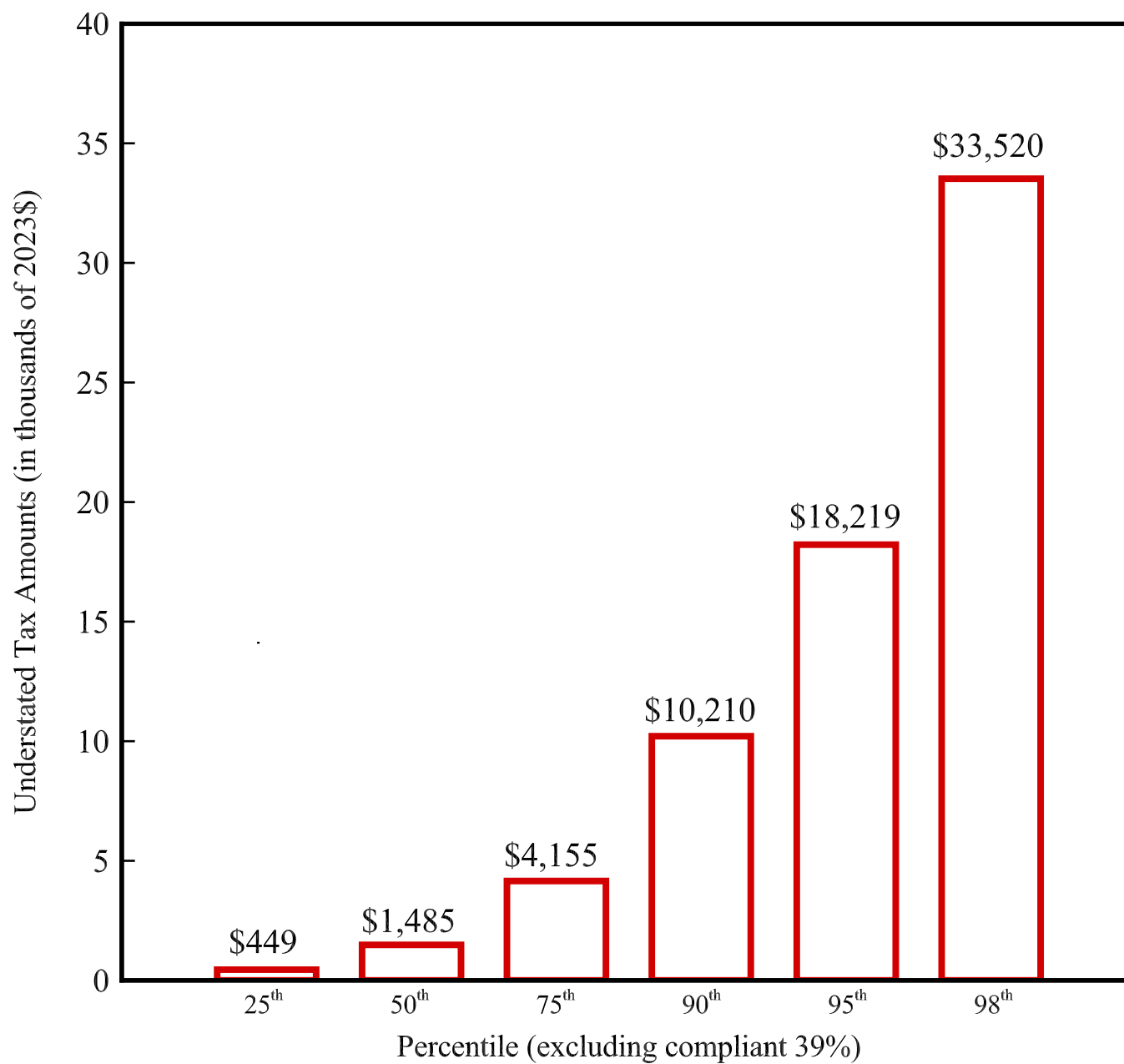


Proprietors Ranked by Understated Taxes



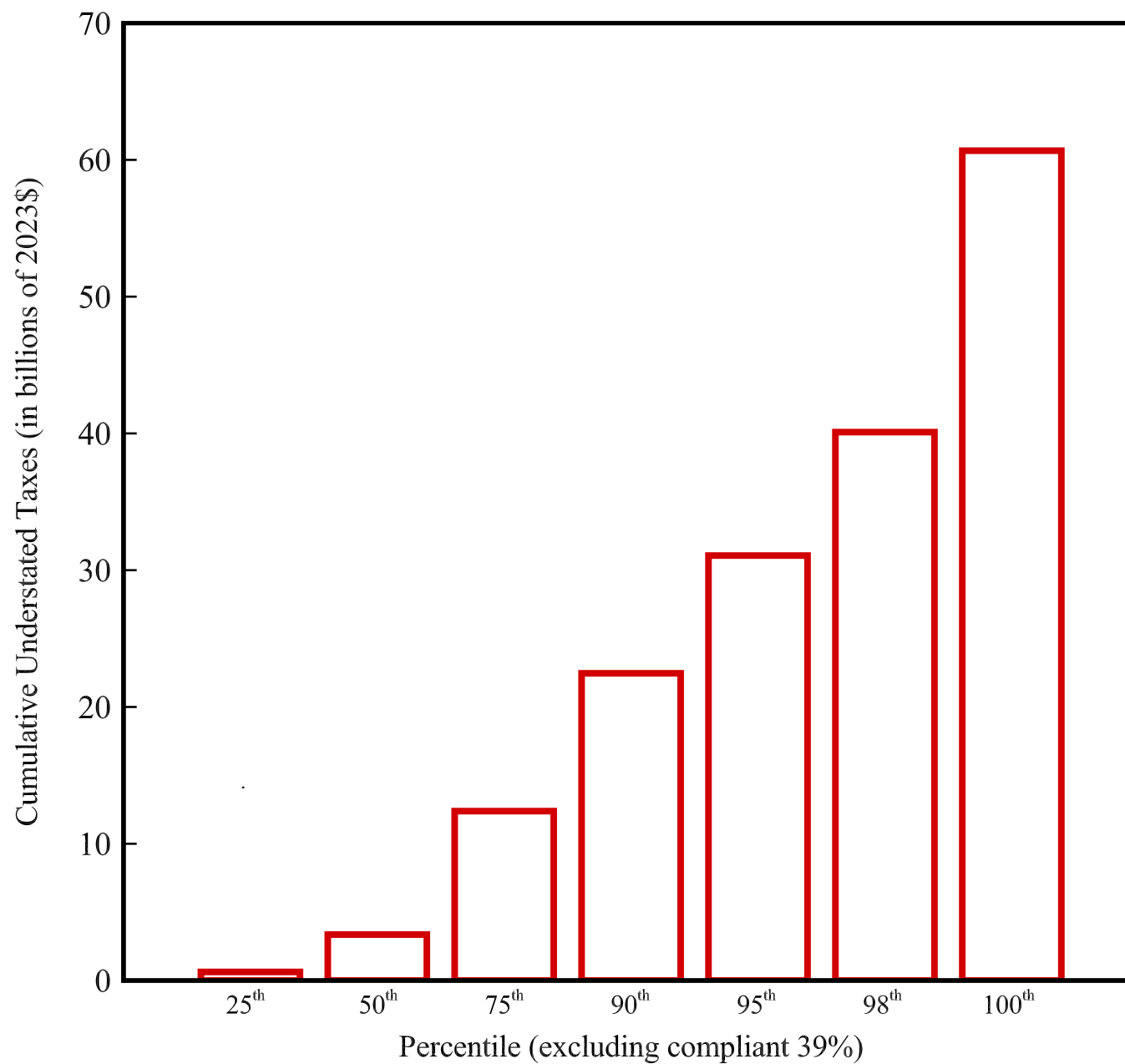


Proprietors Ranked by Understated Taxes





Cumulated Understated Taxes





Nonpecuniary Motives for Compliance

- TAS surveys intended to elicit nonpecuniary motives
- How?
 - Construct samples of sole proprietors
 - Use DIF scores indicating likelihood of audit
 - Group proprietors by DIF score
 - Compare responses of low-DIF and high-DIF groups



TAS Survey Main Results

- Compare lowest and highest compliance groups
- Where similar:
 - Agree tax rules complicated
 - Know consequences of underreporting
 - Profess moral obligation to pay taxes
- Where different:
 - High-compliance more trusting in IRS/govt
 - High-compliance rely more on preparers



Does Evasion Occur Across Income Distribution?

- Evidence from NRP random-audit studies (no DCE)
 - Available publicly only for total incomes
 - Shows underreporting across the distribution
- Estimates of very top depend on DCE adjustments



Shares of Unreported Total Incomes (no DCE)

